Math 443 Bonus HW 3

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Bonus Problem 1

Prove that the inverse of an orthogonal matrix is equal to it's inverse.

Proof. This is a given through the definitions. Take some orthogonal matrix Q. We know by definition of an orthogonal matrix, that $QQ^T = I = Q^TQ$. Furthermore, the inverse of a square matrix is defined as $AA^{-1} = I = A^{-1}A$. Let Q^{-1} be the inverse of Q, assuming it exists, then we would have $QQ^{-1} = I$, but we also have $QQ^T = I$, thus $QQ^{-1} = QQ^T$, which yields us $Q^{-1} = Q^T$ after left multiplying by the transpose. Furthermore, this is guaranteed to exist as it is a necessity that $QQ^T = I$ for the matrix to be orthogonal. $\square$